

Poornachandra G.

Program Lead | AI enablement | Agentic AI Systems

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PROFESSIONAL SUMMARY

Program and engineering leader with 10+ years of experience managing cross-functional technical programs, Agile delivery, and validation/QC frameworks. Hands-on experience building and evaluating Generative AI systems including prompt engineering, RAG pipelines, multi-agent workflows, and LLM output evaluation (accuracy, hallucination, consistency). Experienced using Azure AI Foundry, LangChain, Promptfoo, DeepEval and enterprise reporting tools to develop and validate enterprise GenAI solutions.

SKILLS & TECHNOLOGIES

GenAI Fundamentals	Prompt Engineering • Context Engineering • LLM Output Evaluation • Hallucination Analysis • Fine-tuning • OpenAI APIs • Anthropic APIs • Gemini APIs • LiteLLM • Ollama • AI Playgrounds
Prompt Engineering & PromptOps	Prompt Lifecycle Management • Prompt Versioning • Prompt Registry • Prompt Evaluation • Prompt Optimization • Zero-shot/Few-shot/Role-based/Multi-turn Prompt Design
AI Evaluation & Observability Tools	Promptfoo • DeepEval • Ragas • LangSmith • Langfuse • OpenAI Evals • MLflow
RAG & Knowledge Validation	RAG Pipelines • RAG Evaluation (Ragas) • LangChain • LangGraph • GraphRAG • MCP • Multi-Agent Systems • Azure AI Search • Semantic Search • Embeddings • Retrieval Quality Validation • Golden Dataset Design • Benchmark Query Design
AI Validation & Quality Engineering	Test Case Design (Functional, Regression, Exploratory) • Accuracy Benchmarking • Root Cause Analysis • Acceptance Criteria • LLM-as-a-Judge (LLMaaS) • Responsible AI Validation • Bias & Fairness Assessment • Red Teaming • Latency, Faithfulness & Retrieval Metrics
Program Coordination & Delivery	Program Management • Agile • Scrum • Sprint Planning • Jira • Confluence • SharePoint • Cross-functional Team Coordination • Technical Roadmapping • Stakeholder Management • Meeting Minutes • Executive Reporting • Risk Management • Team Building & Mentoring
Automation, Data & Reporting	Python • SQL • Bash • Excel (Advanced) • Power BI • Streamlit • Plotly • Matplotlib • Seaborn • AWS (EC2, S3, Batch) • Azure (Foundry, AI Search, Blob Storage) • Databricks • Program Tracking • Dashboards • Trackers • Action Item Logs • Risk Registers • Validation Status Reports
Document AI & Enterprise Knowledge	PyMuPDF • Docling • unstructured.io • Microsoft Work IQ
Domain Expertise	10+ years designing and validating data pipelines and QC frameworks in a regulated precision-medicine environment (oncology, clinical genomics)

PROFESSIONAL EXPERIENCE

Cellworks – Lead Clinical Translation | 2024 – Present

- Coordinated program tracking across Bioinformatics, GenAI, and Data Engineering teams, establishing Jira-based sprint cadences, risk registers, and a Power BI dashboard that reported on milestone status, and risk mitigation actions, enabling leadership to make data-driven decisions.
- Architected and deployed enterprise RAG pipelines using Microsoft Azure AI, LangChain, and enterprise document repositories for entity relationship extraction, knowledge discovery, and a HIPAA-compliant patient query engine for external clients; Validated retrieval quality through structured benchmark queries and golden datasets.
- Designed and maintained golden datasets, benchmark queries, and expected-answer sets that powered repeatable evaluation of retrieval quality, response accuracy, hallucination reduction, and regression testing across GenAI applications, achieving a 30% reduction in post-deployment defects.
- Built reusable prompt libraries and test suites, zero-shot, few-shot, role-based, multi-turn, edge-case, and negative scenarios - using Promptfoo and DeepEval to validate GenAI feature accuracy and consistency before release.
- Supported cross-functional Go/No-Go release decisions by defining quality gates, validation criteria, and release readiness metrics tracked in Jira and visualized through Power BI.
- Communicated program strategy, technical progress, risks, and recommendations to executive leadership through executive-ready dashboards and reports, reinforcing stakeholder alignment and securing continued investment.

Cellworks – Senior Scientist, Reporting Manager | 2021 – 2024

- Built and led a multidisciplinary technical team through structured onboarding and mentoring programs, driving sprint-based delivery, code reviews, and quality governance.
- Designed and owned validation and quality-control frameworks, including structured test design, logging & classification of defects, acceptance criteria, and accuracy benchmarking for an Ensembl variant classification model, improving classification accuracy by 40%+.
- Developed Python-based automation for data ingestion, validation, and monitoring workflows for various pharma collaborations including, Takeda and Genentech, reducing manual review effort while improving process reliability.
- Coordinated data intake and validation across multiple external vendors and heterogeneous data sources, standardizing validation criteria and thresholds, and maintaining a risk register that tracked data quality issues and mitigation actions.
- Established program dashboards and recurring executive reports to track validation progress, delivery status, risks, quality trends, release readiness, and recommendations for leadership and stakeholder reviews.
- Implemented reusable documentation, SOPs, and training materials to promote consistent process adoption across teams.
- Drove early adoption of LLM-assisted, RAG-based workflows for oncology standard-of-care recommendations, integrating prompt engineering, knowledge indexing and LLM output evaluation to deliver evidence-based treatment suggestions.

Cellworks – Senior Scientist, Clinical Translation | 2018 – 2021

- Designed, benchmarked, and optimized multiple bioinformatics analysis pipelines to reduce cost without compromising accuracy, performance, or quality, documenting comparative evaluations for technical and business stakeholders.
- Led a cross-institutional collaboration with Cleveland Clinic Florida to develop and validate predictive analytics pipelines using longitudinal clinical datasets, coordinating project milestones, technical deliverables, stakeholder communications, and executive reporting.

- Designed and operationalized a validation framework for a pediatric clinical cases category, establishing acceptance criteria, quality checkpoints, and interpretation guidelines that standardized output quality and accelerated clinical decision support.

Cellworks – Scientist, Clinical Translation | 2017 – 2018

- Built and curated reusable clinical knowledge and reference libraries, reducing case turnaround time by 20%+ through standardized knowledge reuse and process optimization, and maintained dashboards to track knowledge usage.
- Evaluated and benchmarked multiple external analysis platforms for accuracy, reproducibility, and reliability, presenting comparative findings and recommendations to technical stakeholders and incorporating risk register updates.
- Designed a rule-based variant classification and prioritization framework to standardize output quality, improve report consistency, and streamline downstream clinical interpretation, and documented validation criteria.

Cellworks – Associate Scientist | 2015 – 2017

- Standardized and automated large-scale data harmonization and ETL workflows across 500+ clinical cases and 10,000+ external records, improving data quality and processing consistency.
- Coordinated data standardization initiatives for data from multiple external partner institutions, ensuring consistent data formats, validation rules, and integration requirements.
- Automated recurring report-generation workflows using scripting and standardized templates, reducing weekly reporting effort from ~15 hours to minutes while improving reporting accuracy and consistency, and publishing executive-ready reports.

AI & INDEPENDENT PROJECTS

- **SQL Chat:** Designed a multi-agent system that translates natural-language questions into executable SQL, automatically runs statistical validation checks, and produces stakeholder-ready summaries. - demonstrates prompt engineering, search/query validation, grounding, validation and stakeholder-ready reporting.
- **PathoParse:** Built an LLM-assisted, rule-based extraction of structured data from unstructured pathology reports, with validation logic for extraction accuracy - demonstrates unstructured-source data validation.
- **Infervise:** Created a unified interface for discovery, categorization, proxy configuration, and benchmarking of open generative AI APIs (NVIDIA NIM, Mistral, Groq, OpenRouter, Ollama, PublicAI) - demonstrates structured benchmarking and comparison of LLMs across providers.

EDUCATION

B.Tech, Biotechnology - JNTUA

2015

PUBLICATIONS & ABSTRACTS

Co-author on 9 peer-reviewed publications and conference abstracts (J Neurooncol, J Clin Oncol, Blood/ASCO) on computational biomarker modeling and therapy-response prediction, 2018-2021. (Available on ORCID: [0009-0009-1610-1205](https://orcid.org/0009-0009-1610-1205))